

Nicolò Rinaldi

Date of birth: 26/11/1999

Place of birth: Mirandola, Italy

Nationality: Italian

CONTACT

Via della Pace, 71,
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https://nicolo-rinaldi.github.io

nicolo-rinaldi

nicolo-rinaldi **(Github)**

WORK EXPERIENCE

Sease.io London, United Kingdom

Business or Sector Information and communication | **Email** n.rinaldi@sease.io | **Website** https://sease.io/

Software developer/Data Scientist

15/04/2025 – Current

- AI integration and reasearch in search engines
- open source search engines developement
- search consultantancy

Siav S.p.A. Rubano (PD), Italy

Data scientist

04/03/2024 – 04/09/2024

Conducted research in NLP for my Master's thesis, focusing on neural semantic search and information retrieval. Evaluated multilingual pre-trained and fine-tuned embedding models for retrieving relevant documents from Italian datasets, with a focus on their applicability to Retrieval-Augmented Generation (RAG) systems.

Italy

Private tutor

2018 – Current

tutoring activity to middle, high school and occasionally also university students in Mathematics, Physics and Informatics, carried out online via Meet/Zoom or in-person (from 0 up to 15 hours/week, depending on the needing of the students)

EDUCATION AND TRAINING

30/09/2022 – 13/12/2024 Padova, Italy

Master's Degree in Data Science (LM-91) Università degli Studi di Padova

Address Via VIII Febbraio, 2, 35122 , Padova, Italy | **Website** https://www.unipd.it/ | **Field of study** Data Science, Mathematics , Statistics | **Final grade** 110 Cum Laude | **Thesis** Exploring Multilingual Embeddings for Italian Semantic Search: A Pretrained and Fine-tuned Approach

15/09/2018 – 30/09/2022 Bologna

Bachelor's Degree in Mathematics (L-35) Alma Mater Studiorum - Università di Bologna

Address Via Zamboni, 33, 40126, Bologna | **Website** https://www.unibo.it/ | **Field of study** Mathematics | **Final grade** 101/110 | **Thesis** Sulla fattorizzazione CUR mediante interpolazione empirica discreta

PROJECTS

Document enrichment with LLMs

Contributed to Apache Solr by developing and integrating an LLM-based document enrichment pipeline for the indexing process. Implemented a new DocumentEnrichmentUpdateProcessorFactory enabling documents to be enriched at index time by generating structured information from existing fields through configurable LLM prompts. The feature supports multiple input fields, single- and multi-valued output fields, and various data types including text, numeric, boolean, and date fields, with integration through LangChain4j. Developed comprehensive tests, configuration validation, error handling, and technical documentation for the new functionality.

Link https://github.com/apache/solr/pull/4259

Neural transition-based parsing for dependency grammars with Arc-Eager

This project is about neural transition-based parsing for dependency grammars with unlabelled dependencies, with 2 different neural approach: BiLSTM and BERT. Both models are trained and tested on the Irish IDT dependency treebank from the Universal Dependency (UD) project. Finally, the outcomes are evaluated against the relative State of the Art (SotA) results.

Link https://github.com/nicolo-rinaldi/neural-based-dependency-parsing-with-ArcEager

Prediction of smoking status given medical features

According to global statistics, smoking is responsible for millions of deaths each year and is a leading cause of preventable diseases. This project seeks to develop a predictive model that utilizes medical and personal data variables to determine whether an individual has ever smoked or not. By analyzing a dataset comprising health measurements and smoking history information, we aim to identify key indicators that can accurately classify individuals based on their smoking status.

Link https://github.com/nicolo-rinaldi/smoking-prediction

LANGUAGE SKILLS

MOTHER TONGUE(S): Italian

Other language(s):

English

Listening B2
Reading B2
Writing B2
Spoken production B2
Spoken interaction B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Google Drive | Microsoft Word | Zoom | Microsoft Office | Agile development

PROGRAMMING SKILLS

Java (computer programming) | C++ | MATLAB | Python (computer programming)

HOBBIES AND INTERESTS

Hiking/Outdoor activities

Tennis

Japanese culture

VOLUNTEERING

15/09/2025 – CURRENT Mirandola

Cat Shelter

- Administration of treatments (tablets, IV fluids, antiparasitic medications, etc.)
- Kennel cleaning and maintenance
- Cleaning and maintenance of the facility
- Management of new arrivals and communication with prospective adopters

Links https://isoladelvagabondo.it/ | https://www.instagram.com/igattidellisoladelvagabondo/